

Vol 16 — The Substrate Operator Algebra: the linear-algebra representation of BST (chapter draft v0.3 full prose, Lyra, 2026-06-29). The Standard Model’s fermion + gauge + cosmology structure = the spectral invariants of one block-diagonal operator algebra  $A$  on the substrate Hilbert space  $H^2(D_{IV}^5)$ , with a single source of noncommutativity ( $\hat{V}$  off-diagonal = CKM). Built from the weekend linearization F408–F412, closure-verified F422. Consolidating reformulation per the linearization standing order; no new claims, count 9/26.

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## Vol 16 — The Substrate Operator Algebra

### The linear-algebra representation of BST

The one-sentence version (for everyone)

The whole Standard Model — which particles exist, how heavy they are, how they mix, how strong the forces are, the Higgs, even the cosmological constant — is the eigenvalues, determinants, projections, and traces of six matrices acting on one space. Write the six operators, read off their spectral data, and the Standard Model’s structure comes out. That is what “the substrate is linear algebra” means.

A useful picture: imagine the substrate as a single musical instrument. The six operators are the things you can *measure* about it — its resonant frequencies, how its strings are tuned relative to one another, which notes are in tune and which are deliberately detuned. The Standard Model is not a separate score written on top of the instrument; it *is* the list of those measurements. Everything below makes this precise.

### Why an operator algebra at all

BST places all physics on a single bounded symmetric domain,  $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ , and its Hardy space of boundary-regular holomorphic functions,  $H^2(D_{IV}^5)$ . A bounded symmetric domain is scale-free (conformal): it has no preferred length. That single fact has a sharp consequence — every linear operator natural to the domain is *dimensionless*, and therefore every spectral invariant (eigenvalue, ratio of eigenvalues, determinant, trace, Casimir) is a pure number. The Standard Model’s dimensionless content — generation ratios, mixing angles, gauge  $\beta$ -coefficients, the cosmological-constant exponent — is exactly the kind of thing such invariants can produce, and the dimensionful content (the overall mass scale) is exactly the kind of thing they *cannot* (see “Scope and the count” below). The operator-algebra language is therefore not a stylistic choice; it is the honest bookkeeping of what a scale-free domain can and cannot fix.

The space and the algebra

The substrate Hilbert space decomposes into three commuting sectors,

$$H = H_{\text{gen}} \otimes H_{\text{gauge}} \otimes H_{\text{Clifford}},$$

carrying respectively the generation/flavor structure, the gauge structure, and the discrete (spin-statistics, doubling) structure. On  $H$  acts a small operator algebra  $A$ . Its defining structural fact is that it is block-diagonal across the three sectors:

$$[\hat{H}, \hat{C}_A] = [\hat{H}, \hat{Z}] = [\hat{C}_A, \hat{Z}] = 0.$$

The flavor operators, the gauge operators, and the discrete operators all commute with one another. All of the Standard Model's nontrivial *interference* — the thing that makes the theory more than a list of independent numbers — lives inside a single sector and a single off-diagonal operator, as we will see.

The six operators

| operator    | sector                | what it is                             | spectral data → physics                                                                                                                                                                  |
|-------------|-----------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\hat{H}$   | $H_{\text{gen}}$      | conformal-weight (SO(2) generator)     | eigenvalues $\{n_C/2, N_C/2, 0\} = \{5/2, 3/2, 0\}$ = the three generations (Wallach points)                                                                                             |
| $\hat{D}$   | $H_{\text{gen}}$      | fiber-dimension (Korányi–Wolf stratum) | eigenvalues $\{n_C, \text{rank}, 0\} = \{5, 2, 0\}$ = the support strata                                                                                                                 |
| $\hat{V}$   | $H_{\text{gen}}$      | weak↔mass rotation                     | off-diagonal; the CKM matrix (dual- $\rho$ angles)                                                                                                                                       |
| $\hat{C}_A$ | $H_{\text{gauge}}$    | Casimir on the gauge adjoints          | eigenvalues $h^\vee = \{N_C, \text{rank}\} = \{3, 2\}$ = SU(3) <sub>C</sub> , SU(2) <sub>L</sub> dual-Coxeter numbers                                                                    |
| $\hat{Z}$   | $H_{\text{Clifford}}$ | $Z_2$ Shilov involution                | $\hat{Z}^2 = I$ , spectrum $\{\pm 1\}$ ; Clifford module dim $2^{N_C} = 8$                                                                                                               |
| $\hat{P}_0$ | $H_{\text{gen}}$      | Higgs/VEV projection                   | projection onto the $\nu=0$ eigenspace of $\hat{H}$ ; Yukawas are its matrix elements; $y_t=1 \nRightarrow \text{top} \in \ker(\hat{H})$ ; EW breaking = $\langle \hat{P}_0 \rangle = v$ |

Two of these,  $\hat{H}$  and  $\hat{D}$ , are simultaneously diagonal ( $[\hat{H}, \hat{D}]=0$ ), and their common eigenbasis is the mass eigenbasis. The remaining structure is read off sector by sector.

The masses (sector  $H_{\text{gen}}$ ): characteristic polynomials and determinants

The generation sector is where the fermion mass hierarchy lives, and it is built from two diagonal operators and one function evaluated on their spectrum.

Charged leptons and down-type quarks (bulk deposit). These masses are the characteristic-polynomial value  $d(\hat{H})$  — the Harish-Chandra formal degree

$$d(\nu) = (5/2 - \nu)(1 - \nu)(2 - \nu)(3 - \nu)(4 - \nu),$$

evaluated on the generation spectrum, giving  $d(\hat{H}) = \text{diag}(0, -15/16, 60)$ . The two nontrivial values factor cleanly into substrate integers:  $|d(\mu)| = N_C \cdot n_C / \text{rank}^4 = 15/16$  and  $|d(\tau)| = N_C \cdot n_C \cdot \text{rank}^2 = 60$  (F390). The ratio of consecutive generations is  $\text{rank}^6$  — a pure power of the rank, with no fitted weight.

Up-type quarks (boundary concentration). Here the Yukawa is a determinant of the fiber operator,  $\mu^{\hat{D}} = \text{diag}(\mu^5, \mu^2, 1)$ , with  $\mu = 1/\text{rank}^3$  (F392, F406). The top quark sits at  $\hat{D}=0$ , where the determinant saturates at 1, forcing  $y_t = 1$  and hence  $m_t = v/\sqrt{2} \approx 173.9$  GeV (F373, F387). The top mass is therefore not a fitted Yukawa but the statement that the top occupies the kernel of the conformal-weight operator.

The two fermion families differ by *which* spectral functional reads the same  $\nu$ -ladder: a characteristic polynomial for the bulk (down/lepton) deposit, a determinant for the boundary (up) concentration. One ladder, two readings.

The mixing (sector H\_gen): the single source of noncommutativity

The operator  $\hat{V}$  rotates from the mass eigenbasis to the weak eigenbasis; it is built from the dual- $\rho$  angles (the Cabibbo angle  $\theta_{12}$ , the angle  $\theta_{23} = \psi = \arctan(N_c/n_c)$ , and  $\theta_{13}$ ). Its physical content is a non-vanishing commutator:

$[\hat{H}, \hat{V}] \neq 0$  CKM mixing exists.

This is the heart of the representation. The single source of noncommutativity in all of A is that  $\hat{V}$  is off-diagonal in the mass eigenbasis. Because the mass operators  $\hat{H}$  and  $\hat{D}$  are simultaneously diagonal,  $\hat{V}$  fails to commute with *both* of them — but  $[\hat{H}, \hat{V}] \neq 0$  and  $[\hat{D}, \hat{V}] \neq 0$  are the *same* phenomenon (the misalignment of the mass and weak bases), not two independent facts. The three sectors of A still commute with one another; the only place the algebra is non-abelian is this single off-diagonal rotation. If  $\hat{V}$  were diagonal, the mass and weak bases would coincide and there would be no mixing at all. The misalignment direction is the conformal- $\rho$  direction (F376, F384), and the structure constant carries it explicitly:

$$[\hat{H}, \hat{V}]_{23} / \hat{V}_{23} = \nu_2 - \nu_3 = N_c/2 = \rho_2.$$

(Self-correction, retained from v0.1: an earlier phrasing called this “the single nontrivial commutator  $[\hat{H}, \hat{V}]$ ”; since  $[\hat{D}, \hat{V}] \neq 0$  as well, the precise statement is “a single source of noncommutativity —  $\hat{V}$  off-diagonal — manifesting in both commutators.”)

The gauge sector (H\_gauge): Casimir eigenvalues

The gauge structure is the spectrum of one Casimir operator. The dual-Coxeter numbers of the gauge groups are its eigenvalues on the adjoint representations:  $h^{\vee}(\text{SU}(3)_c) = N_c$  and  $h^{\vee}(\text{SU}(2)_L) = \text{rank}$  (F399, F405). The QCD one-loop  $\beta$ -coefficient is then a linear combination of this spectrum:

$$b_3 = (11 \cdot \hat{C}_A - 2 \cdot n_f)/3 = (11 N_c - 2 C_2)/3 = g,$$

with the fermion count  $n_f = \text{rank}(\text{rank}+1) = C_2$  forced by the matter sitting in the  $\text{SU}(2)_L$  fundamental (dim of the fundamental of  $\text{SU}(N)$  equals  $h^{\vee} = N$ , and  $h^{\vee}(\text{SU}(2)) = \text{rank}$ ). The couplings themselves *run* — their boundary conditions are open — but the  $\beta$ -coefficients are fixed integers read off the Casimir spectrum. Notably,  $b_3 = g$  is an *output*: the integer  $g$  is not inserted, it falls out of the combination, which is the strongest form of target-innocence (F417 tier A).

The discrete sector (H\_Clifford): the involution and the cosmological constant

The operator  $\hat{Z}$  is the  $Z_2$  of the Shilov boundary  $(S^4 \times S^1)/Z_2$ ; it squares to the identity, has spectrum  $\{\pm 1\}$ , and its Clifford module has dimension  $2^{\{N_c\}} = 8$  (the  $\text{SO}(7)$  Dirac spinor, F369). The cosmological constant is a trace:

$$\Lambda = \exp(-\text{tr}) = \exp(-2^{\{N_c\}} \cdot n_c \cdot g) = \exp(-280) \text{ (F406)},$$

with the prefactor’s factor-of-2 the single  $Z_2$  quotient (F404). The same involution  $\hat{Z}$  is responsible for the muon’s factor “2”, for spin-statistics, and for the  $\text{SO}(2)$  periodic “tick” (the last is an internal register quantity, Cal #50).

The dynamics

The commitment-density flow is a one-parameter semigroup generated by the Casimir of the maximal compact K:

$\rho(\tau) = \exp(-\tau H_B/\hbar)$ ,  $H_B = \text{Casimir}(K)$ ,  $\hat{H}$  = its  $SO(2)$  part.

Because  $\text{spec}(H_B) \geq 0$ , the flow is a *contraction* semigroup — and that single positivity statement is the arrow of time as an operator fact. As  $\tau \rightarrow \infty$  the flow projects onto the  $\nu=0$  minimal representation (the heaviest generation); the mass eigenstates are stationary, while the mixing operator  $\hat{V}$  evolves with relative phases  $e^{-\tau(\nu_i - \nu_j)}$ , which is generation interference.

The closure check: the reformulation hides no prediction (F422)

A consolidating reformulation earns its keep only if it is *faithful* — closing  $A$  under commutation must not secretly generate a new relation among the substrate integers. We verified this directly (F422, symbolic computation, exact rationals):

1. Jacobi is automatic. Because the two mass operators are co-diagonal ( $[\hat{H}, \hat{D}] = 0$ ), the Jacobi term  $[\hat{V}, [\hat{H}, \hat{D}]]$  vanishes identically and  $[\hat{H}, [\hat{D}, \hat{V}]] = [\hat{D}, [\hat{H}, \hat{V}]]$  holds for *any*  $\hat{V}$ . No constraint is imposed.
2. No new invariant. The algebra generated by  $\{\hat{H}, \hat{D}, \hat{V}\}$  with a generic off-diagonal  $\hat{V}$  is the full  $\mathfrak{gl}(3)$  on  $H_{\text{gen}}$ , yet every *diagonal* operator the closure produces lies in  $\text{span}\{I, \hat{H}, \hat{D}, \hat{H}^2\}$  — no new conserved spectral quantity appears outside the banked spectrum  $\{5/2, 3/2, 0\}$ .
3. The structure constants are the spectral gaps.  $[\hat{H}, \hat{V}]_{ij}/\hat{V}_{ij} = \nu_i - \nu_j$  yields  $\{1, 5/2, 3/2\} = \{(n_C - N_c)/\text{rank}, n_C/\text{rank}, N_c/\text{rank}\}$ , all functions of the already-banked spectrum.

So the bracket structure forces nothing new:  $A$  is a faithful reformulation, not a disguised prediction. (This closes the Cal #468 bar under the Cal #472 condition.)

The summary table — Standard Model = spectral invariants

| SM quantity                        | spectral invariant of $A$                                                                                                      |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 3 generations                      | eigenvalues of $\hat{H}$                                                                                                       |
| support strata                     | eigenspaces of $(\hat{H}, \hat{D})$                                                                                            |
| down/lepton masses                 | characteristic polynomial $d(\hat{H})$                                                                                         |
| up-quark masses                    | determinant $\mu^{\{\hat{D}\}}$                                                                                                |
| CKM mixing                         | noncommutativity of $\hat{V}$ (off-diagonal) with the diagonal mass sector ( $[\hat{H}, \hat{V}], [\hat{D}, \hat{V}] \neq 0$ ) |
| gauge groups' $h^{\wedge}v$        | Casimir eigenvalues $\hat{C}_A$                                                                                                |
| QCD $\beta$ -coefficient $b_3 = g$ | $(11 \cdot \hat{C}_A - 2 \cdot n_f)/3$                                                                                         |
| spin-statistics, doublings         | $\hat{Z}$ involution, Clifford $\dim 2^{\{N_c\}}$                                                                              |
| cosmological constant $\Lambda$    | $\exp(-\text{trace})$ of the instanton operator                                                                                |
| arrow of time                      | positivity $\text{spec}(H_B) \geq 0$                                                                                           |

Scope and the count (honest)

This chapter is a consolidating reformulation of established results — no new claims, no count motion — now verified faithful (F422). Its honesty rests on one structural fact and one open frontier.

The structural fact: because  $A$  acts on a scale-free domain, every invariant it produces is dimensionless. This *forces* the four pure scales of physics — the Higgs vev,  $\Lambda$ , Newton's  $G$ , and the neutrino scale — to remain open, since a dimensionless algebra cannot manufacture a dimensionful number from nothing. This is a principled floor, not a ceiling (Cal #469/#470): it is the minimum that must stay open, not a statement that everything else is closed.

The open frontier: the dimensionless sector is not closed. The CKM magnitudes  $V_{cb}$ ,  $V_{ub}$ , the CP phase  $\delta$ , the PMNS angles, and the gauge-coupling ratios are all dimensionless, open, and reachable in principle; and the electron-to-Planck ratio  $m_e/m_P$  joins them if the boundary-to-bulk  $\alpha$ -suppression is derived rather than identified (the “why- $\alpha$ ” program; the count moves only if that boundary overlap is *computed*, not asserted). The

count is therefore a floor with a live dimensionless frontier — currently 9 reachable invariants banked out of 26, held honestly.

One block-diagonal operator algebra, one source of noncommutativity ( $\hat{V}$  off-diagonal = the mixing), and the Standard Model is its spectral data. That is the linear-algebra representation of BST.